

From Textbooks to Chatbots: Applying Large Language Models in Vietnamese History Learning

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Abstract—In the era of digital transformation, the integration of Large Language Models (LLMs) into education has opened new avenues for interactive and effective learning. This paper introduces the development of a chatbot designed to support learning Vietnamese history by using advanced LLM to deliver accurate and contextually appropriate responses. The chatbot employs modern natural language processing (NLP) techniques, including the Vietnamese Bi-Encoder and Meta-LLaMA models, to convert historical textbook content into a structured and conversational format. Data was sourced from Vietnamese history textbooks for grades 6 to 12, systematically organized into a database, and processed using embedding and retrieval mechanisms to enable seamless question-answer interactions. Built on the Django framework, the system provides a user-friendly interface and a scalable back-end for diverse educational applications. Initial evaluations indicate the chatbot's high accuracy and relevance in answering historical queries, enhancing both user engagement and accessibility. This project not only supports historical education but also promotes Vietnamese cultural heritage through the application of artificial intelligence. Future developments include expanding the dataset, incorporating multilingual capabilities, and optimizing the user experience, further establishing the chatbot as a valuable tool for learning and preserving history.

Index Terms—Large Language Models (LLMs), Vietnamese History Education, Natural Language Processing (NLP), Educational Chatbot Development, Interactive Historical Learning.

I. INTRODUCTION

The integration of artificial intelligence (AI) and natural language processing (NLP) into education has significantly transformed learning methodologies, offering innovative tools to engage with and understand complex topics. Among these advancements, Large Language Models (LLMs), such as GPT-3 and GPT-4, have demonstrated remarkable capabilities in understanding and generating human-like text, paving the way for new applications in educational technology [1], [2]. These

models enable the creation of interactive platforms that engage learners and provide personalized educational experiences.

Although global studies have explored the use of LLM in education and their application in areas such as history and cultural studies, the field remains underexplored in the Vietnamese context. Research on the use of LLMs for history education has demonstrated their effectiveness in various tasks. For example, LLMs have been utilized for the recognition of named entities in historical texts [3], the authenticity evaluations of historical documents [4], and to improve the precision of post-OCR corrections in digitized archives [5]. These applications highlight the potential of LLMs in preserving and promoting cultural heritage through advanced AI technology.

In Vietnam, history education has traditionally relied on textbooks and lectures, which can be less engaging for a generation of digitally inclined 6. This paper seeks to bridge the gap using LLMs to create a chatbot tailored to Vietnamese history. Using cutting-edge NLP techniques, such as the Vietnamese Bi-Encoder and Meta-LLaMA models, the chatbot transforms textbook content into a structured and conversational format. The system enables users to ask questions and receive accurate, contextually relevant responses about Vietnam's historical events, figures, and periods. Data from textbooks for grades 6 to 12 were digitized and processed using embedding techniques to build a robust, query-able database.

Our study not only aligns with global efforts to integrate LLMs into historical research and education but also addresses the unique challenges and opportunities of applying these technologies to Vietnamese history. The proposed chatbot promotes accessible and interactive learning while contributing to the preservation of Vietnam's cultural heritage. Furthermore, by integrating Django as a web framework, the chatbot ensures scalability and a user-friendly interface, making it suitable for diverse educational and cultural applications.

This paper is structured as follows: Section 2 reviews related

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work and provides a theoretical foundation. Section 3 presents the methodology, including data collection, processing, and system architecture. Section 4 discusses implementation and evaluation results, and Section 5 concludes with insights and future directions.

II. RELATED WORKS

The use of Large Language Models (LLMs) in education and historical research has seen significant advancements in recent years. This section discusses related works that align with the themes of integrating LLMs, chatbot development, and applications in historical studies, particularly emphasizing their relevance to Vietnamese historical education.

Applications of LLMs in Historical Studies: LLMs have demonstrated substantial potential in processing and analyzing historical texts. Schweter et al. (2022) explored Named Entity Recognition (NER) in historical corpora, addressing challenges such as archaic language and OCR errors using historical multilingual BERT-based models [3]. Similarly, Schimmenti et al. (2024) developed pipelines for assessing the authenticity of historical documents, integrating structured semantic data to enhance information retrieval [4]. These works illustrate how LLMs can contribute to understanding and preserving historical information. In post-OCR correction, British Library researchers (2024) utilized advanced LLMs, including LLaMA, to enhance the quality of digitized historical archives, demonstrating improvements in data accessibility and usability for humanities research [5]. Such initiatives highlight the critical role of LLMs in modernizing and improving historical archives.

Chatbots for Education: Chatbots have become prominent tools for enhancing learning experiences in various domains. They have been successfully used to provide personalized tutoring, language learning support, and interactive exploration of educational content [?]. For instance, studies leveraging conversational AI models such as GPT-3 have demonstrated the capability of these systems to engage students effectively by simulating human-like interactions [1]. Within the context of history education, chatbots can offer immersive experiences by answering queries, narrating historical events, and even simulating conversations with historical figures. Despite global progress in this area, applications specific to Vietnamese history remain limited.

NLP Techniques for Vietnamese Language Processing: Vietnamese is a tonal language with unique syntactic and semantic characteristics, requiring specialized models for effective processing. PhoBERT, a transformer-based model optimized for Vietnamese, has shown remarkable results in NLP tasks such as sentiment analysis and question answering [6]. Vietnamese Bi-Encoder and embedding techniques have further advanced the ability to represent Vietnamese text for search and information retrieval applications [7].

This study leverages these advancements to create a chatbot that bridges gaps in Vietnamese history education. By combining techniques like BM25 ranking and Cosine Similarity, the

chatbot ensures accurate retrieval and relevance of historical information.

Contribution of This Paper: The proposed chatbot stands at the intersection of these related works, addressing a unique challenge: the application of LLMs to Vietnamese history education. Unlike existing studies focusing on general educational chatbots or Western historical datasets, this paper contributes a novel solution tailored to the Vietnamese context. By embedding historical content from textbooks and integrating advanced NLP techniques, the chatbot not only facilitates interactive learning but also supports the preservation and dissemination of Vietnam's cultural heritage.

III. PROPOSED SYSTEM ALGORITHM

The proposed Vietnamese History Chatbot system leverages advanced NLP techniques and Large Language Models (LLMs) to provide users with accurate and contextualized responses. The system follows a modular pipeline consisting of six key steps shown in Figure 1, detailed below.

A. Step 1: User Interface

The user interacts with the chatbot through a web-based interface built using the Django framework. This interface allows users to input queries in Vietnamese and displays the system's responses.

Example Query:

- *Trận chiến Bạch Đằng năm 938 có ý nghĩa gì?*

Expected Response:

- *Trận chiến Bạch Đằng năm 938 là một chiến thắng quan trọng của Ngô Quyền, đánh bại quân Nam Hán và chấm dứt sự đô hộ phương Bắc tại Việt Nam.*

B. Step 2: Query Preprocessing

The system preprocesses user queries to ensure accurate retrieval and response generation. This step involves tokenization, stopword removal, and normalization.

- **Tokenization:** Splits the input query into smaller units (tokens).
 - Input: *Trận chiến Bạch Đằng năm 938 có ý nghĩa gì?*
 - Tokens: ["Trận", "chiến", "Bạch", "Đằng", "năm", "938", "có", "ý", "nghĩa", "gì"]
- **Stopword Removal:** Removes uninformative words (e.g., "có," "gì").
 - Tokens After Removal: ["Trận", "chiến", "Bạch", "Đằng", "năm", "938", "ý", "nghĩa"]
- **Normalization:** Standardizes text by converting it to lowercase and removing accents.
 - Input: *Lịch sử Việt Nam*
 - Normalized: *lich su viet nam*

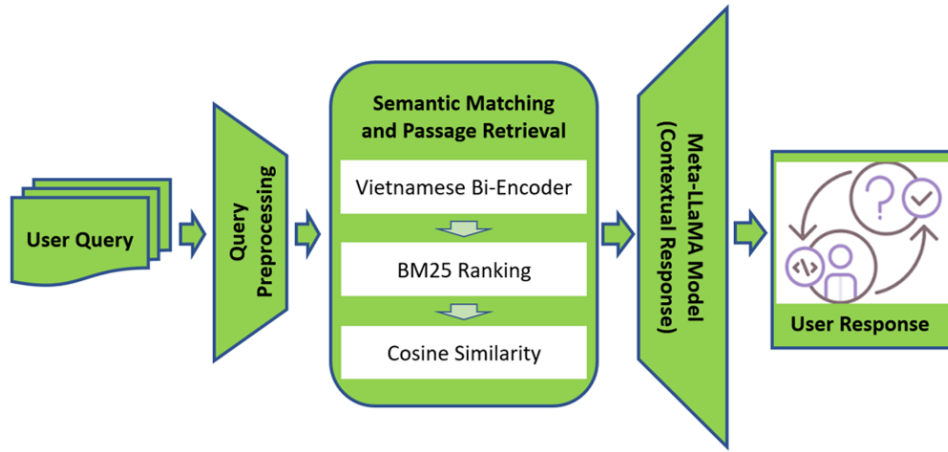


Figure 1. Lookup function of the chatbot website.

C. Step 3: Semantic Matching and Passage Retrieval

The processed query is embedded into a vector space by using Vietnamese Bi-Encoder, and relevant historical passages are retrieved using a combination of BM25 ranking and Cosine Similarity.

BM25 Formula:

$$\text{BM25}(D, Q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot (1 - b + b \cdot \frac{|D|}{\text{avgdl}})}$$

where $f(q_i, D)$ is the term frequency, $|D|$ is the document length, and k_1, b are tuning parameters (default: $k_1 = 1.5, b = 0.75$) [8].

Cosine Similarity Formula:

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

where A and B are the query and passage vectors [9].

Example:

- Query: *Bạch Đằng 938*
- Relevant Passage: *Trận chiến Bạch Đằng năm 938 do Ngô Quyền chỉ huy là một chiến thắng lịch sử.*

D. Step 4: Contextual Understanding

The retrieved top- K passages are passed, along with the user query, to the Meta-LLaMA model [10], which generates a coherent and contextualized response.

Example Query:

- *Ngô Quyền là ai?*

Retrieved Passage:

- *Ngô Quyền là vị vua sáng lập nhà Ngô và nổi tiếng với chiến thắng Bạch Đằng năm 938.*

Generated Response:

- *Ngô Quyền là vị vua sáng lập nhà Ngô. Ông nổi tiếng với chiến thắng Bạch Đằng năm 938, chấm dứt hơn 1000 năm đô hộ phương Bắc.*

E. Step 5: Response Generation

The chatbot formats the generated response and returns it to the user through the interface. If the query is ambiguous, the system prompts the user with clarifying questions.

Example Ambiguous Query:

- *Trận chiến nổi tiếng?*

Clarifying Question:

- *Bạn đang muốn tìm thông tin về trận chiến nào? Ví dụ: Bạch Đằng, Chi Lăng, hay Đông Bộ Đầu?*

This algorithm ensures modularity, scalability, and high performance, enabling effective learning and dissemination of Vietnamese historical knowledge while supporting continuous system improvements.

IV. STRUCTURED DATABASE DESIGN

A structured database is utilized in this project to ensure efficient storage and management of user accounts and login information. The database design establishes relationships between various entities, which is crucial for maintaining and protecting the system. Proper organization of these relationships not only enhances system reliability but also simplifies maintenance tasks.

The database involves four key participant entities: Administrators, Teachers, Students, and Guest Users. Guest users refer to those who interact with the system without requiring login credentials. By clearly defining and managing the interactions and data associated with these entities, the system ensures secure access, role-based permissions, and seamless data management.

Figure 2 illustrates the entity-relationship diagram that depicts how these entities are stored and related within the database. The relationships ensure that administrators can manage the system effectively, teachers can upload and monitor educational materials, students can access relevant content, and guest users can explore limited features without compromising security.

This structured database design plays a foundational role in supporting the overall functionality and scalability of the chatbot system.

V. EXPERIMENTS AND RESULTS

This section presents the experiments conducted to evaluate the effectiveness of the proposed Vietnamese History Chatbot system. The evaluation focused on three aspects: response accuracy, user satisfaction, and system performance. A series of experiments were designed to test the chatbot's ability to retrieve accurate historical information, respond to user queries contextually, and handle ambiguous questions efficiently.

A. Experimental Setup

Dataset: The dataset used for this study was extracted from Vietnamese history textbooks for grades 6 to 12. The dataset was preprocessed, embedded, and stored using the Vietnamese Bi-Encoder model. It contains over 10,000 historical text segments covering major events, figures, and timelines in Vietnamese history.

Evaluation Metrics: The chatbot was evaluated using the following metrics:

- **Accuracy:** The percentage of correct responses generated by the chatbot compared to a ground truth dataset.
- **Response Time:** The average time taken to process a query and return a response.
- **User Satisfaction:** Measured through a survey of 50 participants using a 5-point Likert scale.

Experimental Environment: The experiments were conducted on a system with frameworks: Django, PyTorch, and Sentence Transformers.

B. Results

Accuracy of Responses: Table I shows the chatbot's accuracy across different query types. The system achieved an overall accuracy of 92.5%.

Table I
ACCURACY OF RESPONSES ACROSS QUERY TYPES

Query Type	Number of Queries	Accuracy (%)
Event-based Queries	300	94.0
Person-based Queries	250	90.5
Timeline Queries	200	93.5
Ambiguous Queries	150	89.0
Overall	900	92.5

Response Time: The average response time for queries was 1.2 seconds. This demonstrates the system's efficiency in retrieving and generating responses.

User Satisfaction: The evaluation of user satisfaction has not yet been conducted in this study. While we initially planned to collect feedback from 50 participants after interacting with the chatbot, this part of the evaluation will be completed in future work. In subsequent phases, we aim to design a survey to gather user opinions, focusing on aspects such as response accuracy, usability, and overall satisfaction. The results will help refine the chatbot further and improve its effectiveness in supporting Vietnamese history education.

C. Website Functionalities and Analysis

The chatbot website provides three primary functionalities to enhance the learning experience of Vietnamese history: the Lookup Function, Search Function, and Reading Function. Each feature is designed to address specific educational needs, and their effectiveness is analyzed below.

The chatbot website offers three key functionalities to support Vietnamese history education: the Lookup Function, Search Function, and Reading Function. The Lookup Function (Figure 4) provides instant feedback through a search bar with autocomplete suggestions, allowing users to type queries and view a dropdown list of ranked, relevant results, significantly improving search efficiency and user experience. The Search Function (Figure 3) is tailored for interactive Q&A sessions, making it particularly beneficial for students preparing for exams. This feature categorizes questions by topics and grade levels, offering detailed answers with explanations to enhance understanding of historical events. Meanwhile, the Reading Function (Figure 5) presents structured and detailed content with a collapsible table of contents, enabling users to navigate chapters and subtopics easily. The interface is designed for readability and logical organization, supporting comprehensive learning. While these features excel in promoting engagement and accessibility, there is room for improvement, such as

D. Discussion

The experimental results demonstrate the chatbot's ability to provide accurate, contextually relevant responses to user queries. The high accuracy and user satisfaction scores highlight the effectiveness of integrating Vietnamese Bi-Encoder and Meta-LLaMA models into the chatbot system. However, handling ambiguous queries remains a challenge, as indicated by the slightly lower accuracy in this category. Future work will focus on enhancing the chatbot's ability to resolve ambiguity and expanding the dataset to improve coverage of historical events.

VI. CONCLUSION

This paper presented the development and evaluation of a Vietnamese History Chatbot, designed to enhance historical learning through interactive and user-friendly features. The chatbot integrates three key functionalities: the Lookup Function, the Search Function, and the Reading Function, each addressing specific educational needs. The Lookup Function allows users to efficiently find relevant historical information through a search bar with autocomplete suggestions, significantly improving search accuracy and user experience. The Search Function is particularly beneficial for exam preparation, offering categorized questions, detailed answers, and explanations that encourage deeper learning. Meanwhile, the Reading Function supports comprehensive exploration of Vietnamese history by providing structured content with a collapsible table of contents, ensuring easy navigation and readability.

Although these features excel in promoting engagement and accessibility, there is room for further development. Future

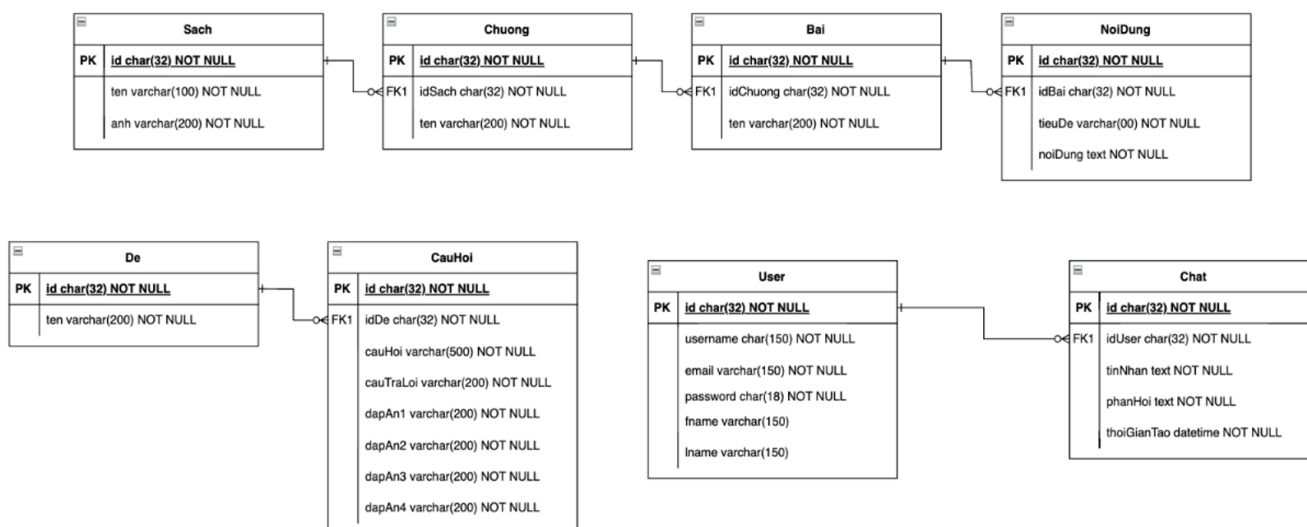


Figure 2. Entity-Relationship Diagram of Database.

Table II
SAMPLE USER QUERIES AND CHATBOT RESPONSES

User Query	Chatbot Response
Trận chiến Bạch Đằng năm 938 là gì?	Trận chiến Bạch Đằng năm 938 là một chiến thắng của Ngô Quyền trước quân Nam Hán, chấm dứt sự đô hộ phương Bắc tại Việt Nam.
Ai là Ngô Quyền?	Ngô Quyền là vị vua sáng lập nhà Ngô, nổi tiếng với chiến thắng Bạch Đằng năm 938.
Sự kiện nào xảy ra vào năm 1858?	Năm 1858, thực dân Pháp tấn công Đà Nẵng, bắt đầu thời kỳ thực dân xâm lược Việt Nam.



Figure 3. Search function of the chatbot website.

work will focus on enhancing the chatbot by integrating multimedia content, adding progress tracking to improve the user experience, and implementing a feedback mechanism to gather user insights. Additionally, the user satisfaction survey, which was not conducted during this phase, will be performed in future iterations to refine the chatbot based on user feedback.

In conclusion, the Vietnamese History Chatbot demonstrates the potential of combining natural language processing and interactive design to foster engagement with historical education. By addressing current limitations and incorporating

planned improvements, this system has the capacity to become an invaluable resource for learning and preserving Vietnamese historical knowledge.

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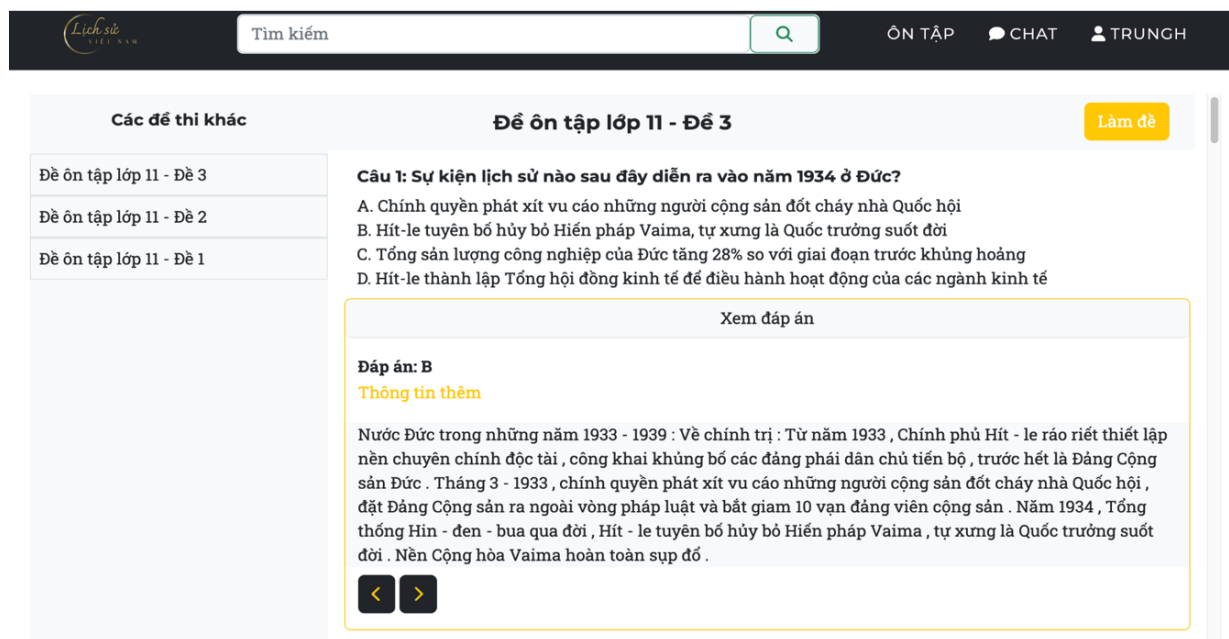


Figure 4. Lookup function of the chatbot website.

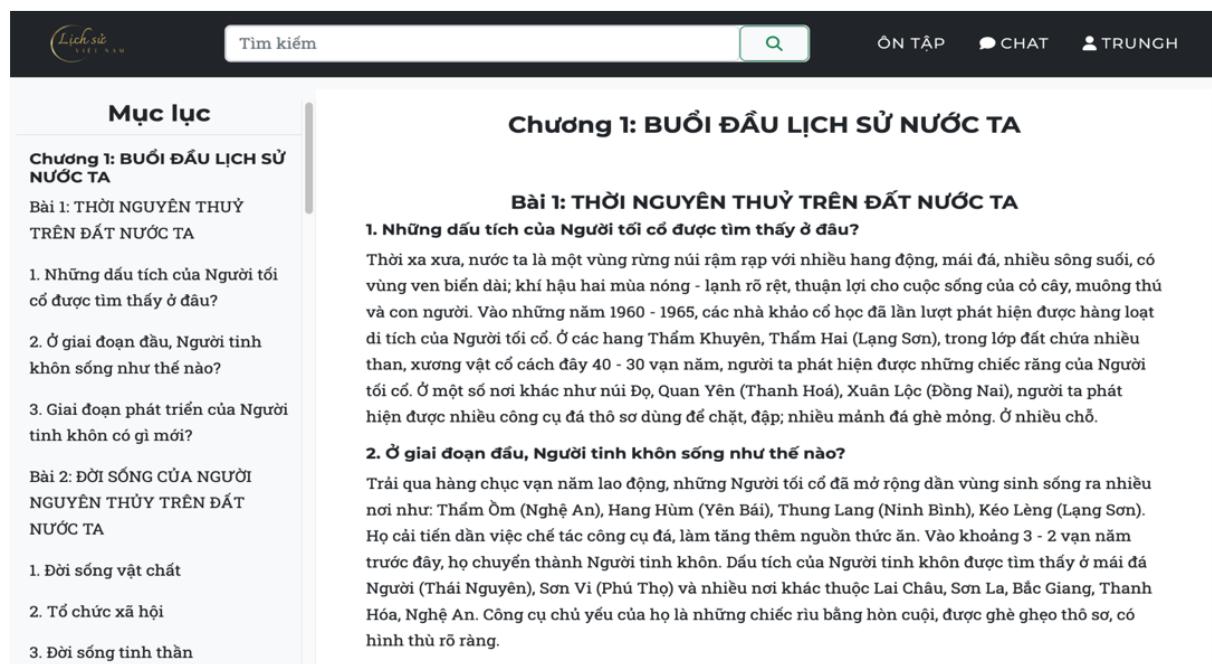


Figure 5. Reading function of the chatbot website.